

Toward Robust Assistive Robot Navigation with EEG Motor-Imagery

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Abstract—Nonverbal human–robot cooperation requires a robot to infer user intent while accounting for uncertainty in the observed cue. Motor-imagery brain–computer interfaces (MI-BCIs) provide a deliberately generated physiological cue for communicating user intent without speech or physical input. We present a decision-point shared-control architecture that filters the output of a participant-specific Riemannian-LDA decoder with unlabeled session alignment and classifier recalibration. The decoder remained fixed during closed-loop navigation. Confidence rejection and multi-window temporal smoothing filter the decoded cue before it can command the robot. Authority passes to the user only at intersection locations where LiDAR sensing has already brought the robot to a stop. Three participants drove a robot through mazes comprising of ten traversals of a four-corner route, yielding 40 corners per participant and 120 corners overall. Under the personalized interfaces, the first accepted command was correct at 118 of 120 corners (98.3%) while 120 of the 124 accepted commands (96.8%) produced successful turn outcomes. The four additional commands represent local recovery overhead following unsuccessful actions. All trials were completed using ROS 2 mobile-manipulator simulation in Gazebo, with an aggregate median decision wait of 0.20 s. The results show that a personalized physiological cue can be integrated with contextual autonomy and local error recovery for cooperative human-robot navigation.

Index Terms—nonverbal cues, motor imagery, brain–computer interface, intention inference, human–robot cooperation, shared control, reject option

I. INTRODUCTION

Nonverbal cues allow robots to infer human internal states and coordinate without relying exclusively on speech. Most computational human-robot interaction (HRI) studies emphasize visible or audible behavior, including gaze, facial expression, gesture, and paralinguistic [1]. Physiological signals provide a complementary channel. EEG motor imagery is covert, hands-free, and deliberately generated to express an intended action. This is critical when physical interfaces are inaccessible to the user, and when the robot must distinguish intended commands from uncertain sensor evidence [2], [3].

Continuous MI control remains difficult because EEG is nonstationary, so decoder accuracy measured within a session need not survive into later ones [10], and transient errors can propagate directly to robot motion. Shared control mitigates this by blending the decoded command with sensor-based autonomy, so that the robot’s own perception of the environment constrains or overrides an erroneous or noisy input rather than letting it drive the platform unmodified [4], [5].

Shared-control systems allocate control authority between the user and the robot in different ways: continuous arbitration blends decoded signal and autonomous controller at every timestep, while discrete handoff lets the user select a goal or waypoint and leaves execution to the robot. Depth-sensor-based shared control now supports both wheelchair mobility and reach-and-grasp assistance for long-term, at-home BCI use [6], which argues for arbitration that keeps the user in the loop without demanding continuous decoding. We adopt the discrete-handoff end and extend it from control allocation to communication. A LiDAR range check governs collision-relevant motion and halts the robot at a junction, and only then does authority transfer to the user, for the single discrete decision of which way to turn. MI is therefore a discrete, context-dependent nonverbal cue whose meaning becomes actionable only when the robot requests a route decision.

This work contributes: (1) a participant-specific Riemannian EEG intent interface with confidence-based rejection; (2) an integrated inference-and-action architecture combining inferred human choices with state-dependent robot autonomy; and (3) an online, three-participant evaluation of decision-to-action behavior.

We describe a participant-specific Riemannian EEG intent interface with confidence-based rejection, connect it to a state-dependent navigation controller, and evaluate the pair online with three participants.

II. INTENT INFERENCE AND COOPERATIVE CONTROL

A. EEG Intent Cue

EEG is acquired from 15 sensorimotor channels of an OpenBCI Cyton–Daisy at 125 Hz. Fig. 1 illustrates the cue-based acquisition methodology together with the 15-channel montage.

We developed participant-specific models from two labeled motor-imagery training sessions. We used FBCSP with LDA for offline characterization of the training data, whereas closed-loop navigation employed a Riemannian covariance representation with LDA. At the beginning of each navigation session, an unlabeled calibration block supported session-specific feature alignment and a classifier recalibration to reduce cross-session variability. No labeled navigation data were required. After this initialization, the model is referred to as the fixed decoder g because its parameters remained unchanged throughout the navigation trial.

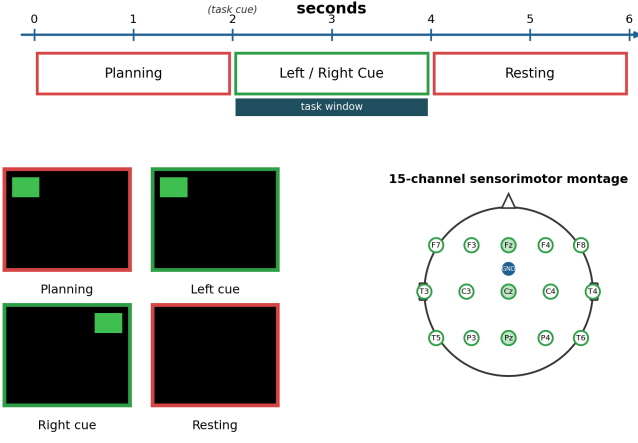


Fig. 1. Cue-based motor-imagery acquisition methodology, adapted from our prior work [8]: trial timing (planning, cue/task window, rest) and the 15-channel sensorimotor montage. LEFT and RIGHT define the two directional MI classes. During navigation, STOP denotes rejected or insufficient evidence rather than a third MI class.

Every 0.2 s, the fixed decoder g maps the latest 2 s EEG window X_t to a rightward discriminant score $s_t = g(X_t)$. A session-dependent confidence reference c_t accommodates score drift without changing the decoder parameters. The decoder output is confidence-gated against c_t ,

$$z_t = s_t - c_t, \quad q_t = \sigma(|z_t|) = \frac{1}{1 + e^{-|z_t|}},$$

$$u_t = \begin{cases} \text{RIGHT}, & z_t > 0, \quad q_t \geq \theta, \\ \text{LEFT}, & z_t \leq 0, \quad q_t \geq \theta, \\ \text{STOP}, & q_t < \theta. \end{cases} \quad (1)$$

where z_t is the score's deviation from the session-dependent reference, $\sigma(\cdot)$ is the logistic sigmoid, $q_t \in [0.5, 1]$ is the resulting confidence value, $\theta \in (0.5, 1)$ is a fixed confidence threshold, and $u_t \in \{\text{LEFT}, \text{RIGHT}, \text{STOP}\}$ is the per-window cue. A cue is issued only when the evidence exceeds the confidence threshold; otherwise, the reject option STOP is reported. A direction is confirmed only after it appears in consecutive decisions, so an isolated misclassification does not affect robot motion. STOP therefore reports insufficient evidence rather than a third directional intention.

B. EEG Preprocessing

Each acquisition window is band-pass filtered into four overlapping mu/beta sub-bands (8–12, 11–15, 14–20, and 20–30 Hz), following the Hamming-windowed FIR design used in our prior offline classification work [8]. For cutoff frequencies f_{low} and f_{high} (normalized by the sampling rate) and filter order N , the impulse response at discrete-time sample index k is

$$h[k] = w_H[k] (2f_{\text{high}} \text{sinc}(2f_{\text{high}}k) - 2f_{\text{low}} \text{sinc}(2f_{\text{low}}k)), \quad (2)$$

with Hamming window $w_H[k] = 0.54 - 0.46 \cos(2\pi k/(N - 1))$. Each sub-band filter is realized in minimum-phase form to limit the group delay introduced ahead of the 0.2 s decision

cadence. Fig. 3 compares the two designs. For the same magnitude specification, the linear-phase equivalent costs roughly 800 ms of group delay across every band. The minimum-phase design keeps delay between about 120 and 340 ms, which is one to two decision intervals instead of four.

C. FBCSP-LDA Characterization

For offline characterization, each band-filtered window $X^{(b)}$ is projected through spatial filters w_j that jointly diagonalize the class-conditional covariances. The normalized log-variance feature is

$$f_j^{(b)} = \log \left(\frac{\text{var}(w_j^\top X^{(b)})}{\sum_{\ell \in \mathcal{J}} \text{var}(w_\ell^\top X^{(b)})} \right), \quad (3)$$

where $\mathcal{J}$ indexes the retained spatial filters. Features are concatenated across bands and classified with shrinkage LDA.

D. Riemannian-LDA Decoding

Closed-loop navigation represents each filter-bank EEG window using regularized bandwidth covariance features within a Riemannian geometry framework [7]. An unlabeled block aligns this representation to the current participant and session, after which the participant-specific LDA is recalibrated once. No labeled navigation samples are used. The resulting fixed decoder supplies the discriminant score $s_t = g(X_t)$ in (1), and its parameters remain unchanged during navigation.

E. Context-Dependent Cooperative Action

The human and robot assume complementary roles. The robot manages continuous motion along a structured route and ignores decoder output during autonomous travel. Forward-range LiDAR sensing identifies a junction, where the robot stops and enters a waiting state. Only then can a confirmed LEFT or RIGHT cue initiate a turn. Autonomous travel resumes after the turn is completed.

The controller state determines when the inferred cue can affect robot motion: the same decoder output is non-actionable during travel but expresses a route choice at a junction. It also makes errors locally recoverable, since a subsequent command can return the robot to the planned route rather than allowing errors to accumulate continuously. Fig. 2 summarizes this cue-to-action pipeline, and Algorithm 1 outlines the corresponding per-window procedure.

The route itself is a four-junction maze, sketched in Fig. 4. Each traversal needs the same RIGHT-LEFT-LEFT-RIGHT sequence. A subject makes four decisions per lap, and any wrong turn is immediately visible as an off-route heading.

III. HUMAN-IN-THE-LOOP EVALUATION

The study was conducted under a protocol approved by the University of Louisville Institutional Review Board (IRB No. 19.0329; approved March 29, 2025). All participants provided written informed consent before participation.

Three participants each completed one continuous closed-loop navigation trial. A trial was ten traversals in a row of the same four-corner route. Each participant therefore encountered

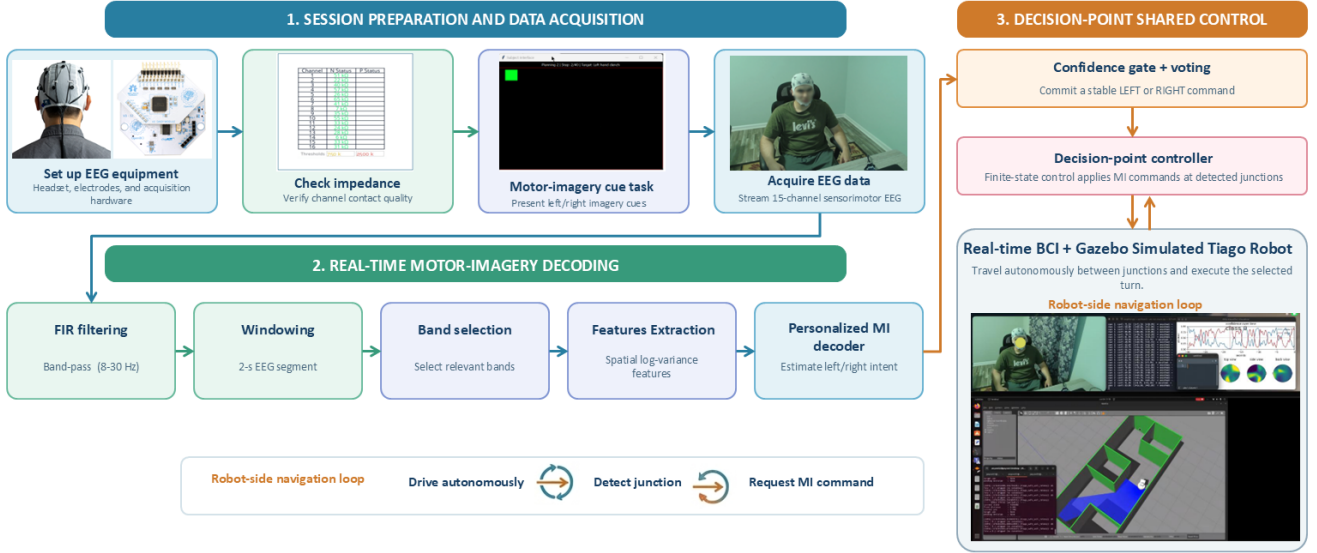


Fig. 2. Closed-loop cooperative-navigation workflow. Filter-bank covariance features undergo unlabeled session alignment and one-time participant-specific LDA recalibration before navigation. The resulting fixed decoder provides an intent estimate to the confidence-rejection and temporal-smoothing stages. A directional cue becomes actionable only after the robot stops at a decision point.

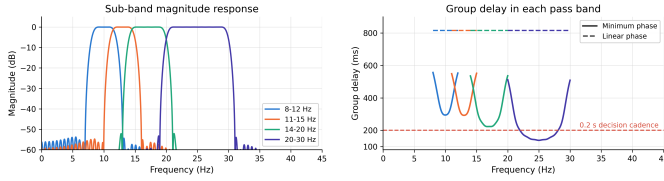


Fig. 3. Filter-bank design. Left: magnitude response of the four mu/beta sub-bands. Right: group delay within each passband for the deployed minimum-phase filters (solid) against linear-phase filters meeting the same specification (dashed), with the 0.2 s decision cadence marked. Minimum phase reduces delay by roughly a factor of three; the residual delay still spans one to two decision intervals in the lower bands.

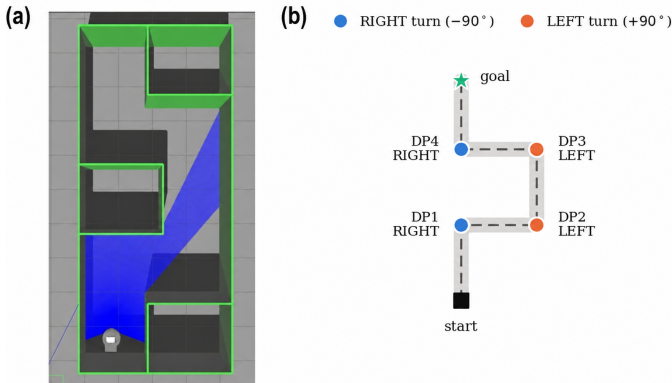


Fig. 4. Navigation environment and required decision sequence. (a) Gazebo maze used for the closed-loop navigation trials. (b) Four decision points (DP1–DP4) in one traversal and the required turn at each junction. Corridor lengths in (b) are schematic.

Algorithm 1 Session-adaptive nonverbal cue inference and decision-point cooperative navigation

Require: EEG stream; participant-specific decoder g ; unlabeled calibration data; confidence threshold θ ; proximity threshold $r_{\min}$; turn tolerance ϵ_{ψ}

- 1: Align the session features and recalibrate g once; hold g fixed thereafter
- 2: Initialize the confidence reference c and state $S \leftarrow \text{AUTO}$
- 3: **loop**
- 4: Preprocess X_t , evaluate $g(X_t)$, and apply (1); temporally confirm the resulting cue u_f
- 5: **if** $S = \text{AUTO}$ **then**
- 6: Drive forward with heading hold and ignore u_f ; if $r_f \leq r_{\min}$, stop and set $S \leftarrow \text{WAIT}$
- 7: **else if** $S = \text{WAIT}$ **then**
- 8: Remain stopped; if $u_f \neq \text{STOP}$, set a left/right 90° target and $S \leftarrow \text{TURN}$
- 9: **else if** $S = \text{TURN}$ **then**
- 10: Rotate toward the target and ignore new cues; if the yaw error is at most ϵ_{ψ} , set $S \leftarrow \text{AUTO}$
- 11: **end if**
- 12: **end loop**

40 decision points, giving 120 decision points across the study. First-command success is an online task-level measure, not offline EEG classification accuracy. At each completed corner, exactly one accepted command produced the successful route outcome. Any additional accepted commands represent recovery overhead following an unsuccessful action.

Table I defines the operational meaning of each cue. Table II reports the navigation outcomes, and Fig. 5 visualizes the corresponding metrics per participant. For $K = 40$ corners, let $D_k \geq 1$ be the number of accepted commands required at corner k . Because each completed corner has one successful

route outcome,

$$M = \sum_{k=1}^K D_k, \quad P_{\text{first}} = \frac{1}{K} \sum_{k=1}^K \mathbb{I}[D_k = 1], \quad (4)$$

$$P_{\text{cmd}} = \frac{K}{M}, \quad E = M - K,$$

where M is the total accepted-command count, P_{first} is first-command success, P_{cmd} is accepted-command success, and E is recovery overhead. Decision wait is reported as the median over all M accepted commands. Pooled results sum K , M , and E across participants and pool their waits before computing the median.

TABLE I
OPERATIONAL SEMANTICS OF THE NONVERBAL CUE.

Robot state	EEG output	Controller response
AUTO	Any output	Ignore the cue and maintain autonomous forward travel.
WAIT	STOP / uncertain	Remain stationary and continue accumulating evidence.
WAIT	Confirmed LEFT / RIGHT	Commit a left/right 90° turn.
TURN	Any new output	Ignore new cues until the committed turn is complete.

TABLE II
RIEMANNIAN–PERSONALIZED COOPERATIVE-NAVIGATION OUTCOMES FOR THREE PARTICIPANTS.

ID	1st cmd correct	Succ. cmds	Extra cmds	Med. wait
S1	40/40 (100%)	40/40 (100%)	0	0.20 s
S2	38/40 (95%)	40/44 (90.9%)	4	0.20 s
S3	40/40 (100%)	40/40 (100%)	0	0.15 s
Aggr.	118/120 (98.3%)	120/124 (96.8%)	4	0.20 s

All three trials were completed. S1 and S3 completed their trials with no additional commands, while S2 required four additional commands across two corners. This shows how task completion can coexist with an imperfect accepted-command sequence.

IV. DISCUSSION

The results illustrate three properties of the architecture. First, the robot evaluates both cue and context rather than mapping every classifier output directly to motion. Second, one-time session alignment and recalibration address cross-session variability, while confidence-based rejection limits weak evidence. Third, the human communicates route preference while the robot supplies sensing, continuous motion, contextual enforcement, and local recovery. The participant- and session-specific Riemannian–LDA decoder remained fixed after its initial recalibration.

Unlike spontaneous gaze or facial behavior, MI is an intentionally generated physiological cue expressing a constrained directional choice. Physical-robot and repeated-session studies with intended users should compare alternative interfaces and

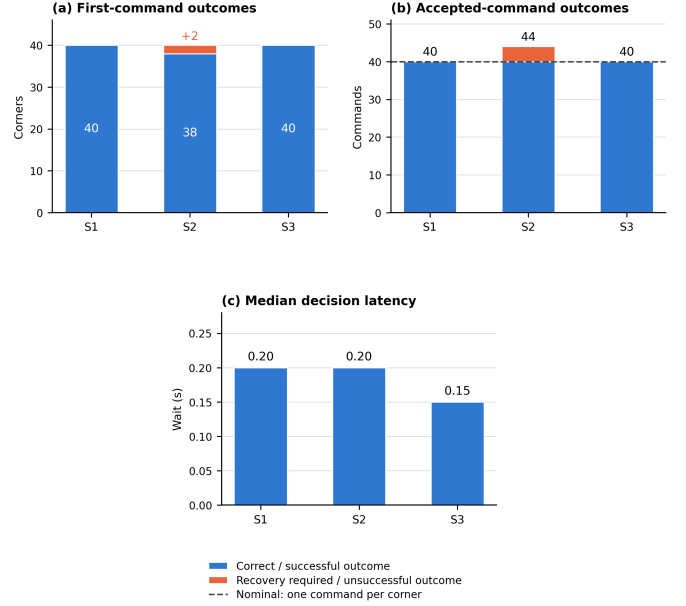


Fig. 5. Per-participant cooperative-navigation outcomes across three continuous 40-decision navigation trials. (a) Corners completed with the first accepted command, with corners requiring recovery stacked above. (b) Successful and unsuccessful accepted-command outcomes relative to the nominal one-command-per-corner requirement. (c) Median online decision wait.

measure calibration burden, workload, and recovery from communication errors.

In this work, EEG motor imagery was used as a nonverbal intent channel within a cooperative navigation loop. The shared-control system completed all three navigation trials and recovered from unsuccessful commands at individual decision points. Application to a physical-robot, extensions to control speed and turn angle, and repeated-session evaluation is the next step toward establishing generalizability of our approach.

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REFERENCES

- [1] J. Urakami and K. Seaborn, “Nonverbal cues in human–robot interaction: A communication studies perspective,” *ACM Transactions on Human–Robot Interaction*, vol. 12, no. 2, pp. 1–21, 2023.
- [2] R. Leeb, L. Tonin, M. Rohm, L. Desideri, T. Carlson, and J. d. R. Millán, “Towards independence: A BCI telepresence robot for people with severe motor disabilities,” *Proceedings of the IEEE*, vol. 103, no. 6, pp. 969–982, 2015.
- [3] L. Tonin and J. d. R. Millán, “Noninvasive brain–machine interfaces for robotic devices,” *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 4, no. 1, pp. 191–214, 2021.
- [4] T. Carlson and J. del R. Millán, “Brain-controlled wheelchairs: A robotic architecture,” *IEEE Robotics & Automation Magazine*, vol. 20, no. 1, pp. 65–73, 2013.
- [5] B. Xu, D. Liu, M. Xue, M. Miao, C. Hu, and A. Song, “Continuous shared control of a mobile robot with brain–computer interface and autonomous navigation for daily assistance,” *Computational and Structural Biotechnology Journal*, vol. 22, pp. 3–16, 2023.

- [6] A. Bellicha, L. Struber, F. Pasteau, V. Juillard, L. Devigne, S. Karakas, S. Chabardes, M. Babel, and G. Charvet, "Depth-sensor-based shared control assistance for mobility and object manipulation: Toward long-term home-use of BCI-controlled assistive robotic devices," *Journal of Neural Engineering*, vol. 22, p. 016045, 2025.
- [7] A. Barachant, S. Bonnet, M. Congedo, and C. Jutten, "Multiclass brain-computer interface classification by Riemannian geometry," *IEEE Transactions on Biomedical Engineering*, vol. 59, no. 4, pp. 920–928, 2012.
- [8] M. A. Hanafy, S. Yusufjonov, D. Yusufjonov, P. Sharafianardakani, M. M. Rayguru, and D. O. Popa, "A robust EEG brain-computer interface approach for decoding hand and foot motor imagery," in *Proceedings of the 18th ACM International Conference on Pervasive Technologies Related to Assistive Environments*, pp. 92–100, 2025.
- [9] H. Ramoser, J. Muller-Gerking, and G. Pfurtscheller, "Optimal spatial filtering of single trial EEG during imagined hand movement," *IEEE Transactions on Rehabilitation Engineering*, vol. 8, no. 4, pp. 441–446, 2000.
- [10] M. A. Hanafy, S. Yusufjonov, P. SharafianArdakani, D. Yusufjonov, M. M. Rayguru, and D. O. Popa, "A dual-validation framework for temporal robustness assessment in brain-computer interfaces for motor imagery," *Technologies*, vol. 13, no. 12, p. 595, 2025.